

Assignment - 1: Data Structure

1. Create a list named 'myList' that is having the following elements: 10,20,30,'apple', True, 8.10.
 - a. Now in the 'myList', append these values: 30,40
 - b. After that reverse the elements of the 'myList' and store that in 'reversedList'

PROGRAM

```
myList=[10,20,30,'apple',True,8.10]
myList.extend([30,40])
print("myList =",myList)
myList.reverse()

reversedList=myList
print("reversedList =",reversedList)
```

OUTPUT

```
myList = [10, 20, 30, 'apple', True, 8.1, 30, 40]
reversedList = [40, 30, 8.1, True, 'apple', 30, 20, 10]
```

2. Create a dictionary with key values as 1,2,3 and the values as 'data','information', and 'text'.
 - a. After that eliminate the 'text' value form the dictionary.
 - b. Add 'features' in the dictionary.
 - c. Fetch the 'data' element from the dictionary and display it in the output.

PROGRAM

```
dict={1:'data',2:'information',3:'text'}
print("Original dictionary =",dict)

dict.pop(3)
print("Text removed dictionary =",dict)

dict[4]="features"
print("Features added dictionary =",dict)

a=dict.get(1)
print("The required element is: ",a)
```

OUTPUT

```
Original dictionary = {1: 'data', 2: 'information', 3: 'text'}
Text removed dictionary = {1: 'data', 2: 'information'}
Features added dictionary = {1: 'data', 2: 'information', 4: 'features'}
The required element is: data
```

3. Create a tuple and add these elements 1,2,3,apple,mango in 'my_tuple'

PROGRAM

```
tup1=(6,20,"ked")
tup2=(1,2,3,'apple','mango')

my_tuple=tup1+tup2
print("The required tuple is =", my_tuple)
```

OUTPUT

```
The required tuple is = (6, 20, 'ked', 1, 2, 3, 'apple', 'mango')
```

4. Create another tuple named `numeric_tuple` consisting of only integer values 10,20,30,40,50

- a. Find the minimum value from the `numeric_tuple`.
- b. Concatenate `my_tuple` with `numeric_tuple` and store the result in `r1`.
- c. Duplicate the tuple named `my_tuple` 2 times and store that in 'newdupli'.

PROGRAM

```
tup1=(6,20,"ked")
tup2=(1,2,3,'apple','mango')
my_tuple=tup1+tup2
print("my_tuple =",my_tuple)

numeric_tuple=(10,20,30,40,50)
print("Minimum value in numeric_tuple =",min(numeric_tuple))

r1=my_tuple+numeric_tuple
print("Concatenated tuple =",r1)

newdupli=my_tuple*2
print(newdupli)
```

OUTPUT

```
my_tuple = (6, 20, 'ked', 1, 2, 3, 'apple', 'mango')
Minimum value in numeric_tuple = 10
Concatenated tuple = (6, 20, 'ked', 1, 2, 3, 'apple', 'mango', 10, 20, 30, 40, 50)
(6, 20, 'ked', 1, 2, 3, 'apple', 'mango', 6, 20, 'ked', 1, 2, 3, 'apple', 'mango')
```

5. Create 2 sets with name `set1` and `set2`, where `set1` contains{1,2,3,4,5} and `set2` to contains{2,3,7,6,1}

- a. Perform the below operation:
 - a. `set1` union `set2`
 - b. `set1` intersection `set2`
 - c. `set1` difference `set2`

PROGRAM

```
set1={1,2,3,4,5}
set2={2,3,7,6,1}

a=set1.union(set2)
print("Union =",a)

b=set1.intersection(set2)
print("Intersection =",b)

c=set1.difference(set2)
print("Difference =",c)
```

OUTPUT

```
Union = {1, 2, 3, 4, 5, 6, 7}
Intersection = {1, 2, 3}
Difference = {4, 5}
```
